

INTRODUCTION TO ARTIFICIAL INTELLIGENCE

Ashenafi.
W(MSC.)

Details about the Course

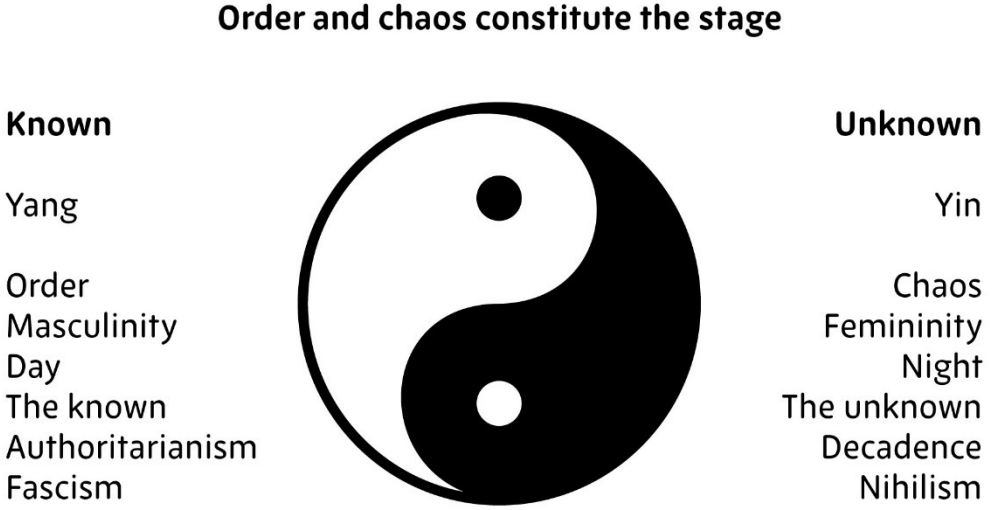
- Course has 5~6 chapters
- The instruction will consist of lectures and lab classes
- Assessment:
    - Continuous Assessments: 60%
    - Final Exam: 40%

- By the end of this course:
    - Describe the key components of the artificial intelligence (AI) field
    - Outline search strategies and solve problems by applying a suitable search method
    - Grasp how agents' reasoning works
    - Understand the issues related to agent planning, handling uncertainty, learning from observation and communicate.

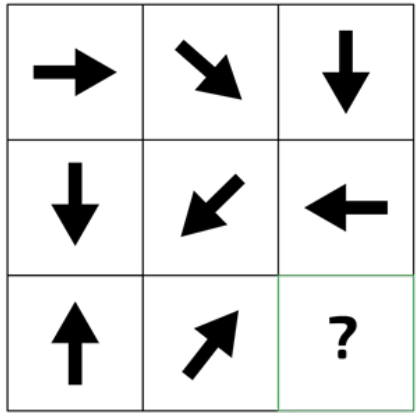
Important Stuff to take care of before beginning the Course

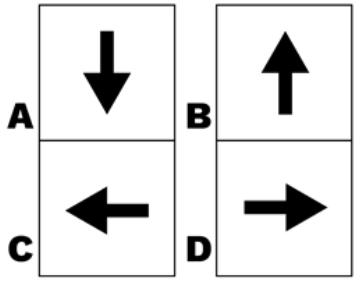
- Representative for the course ?

- Contact:
  - Phone Number:
  - Email:



Intelligence – What is it?





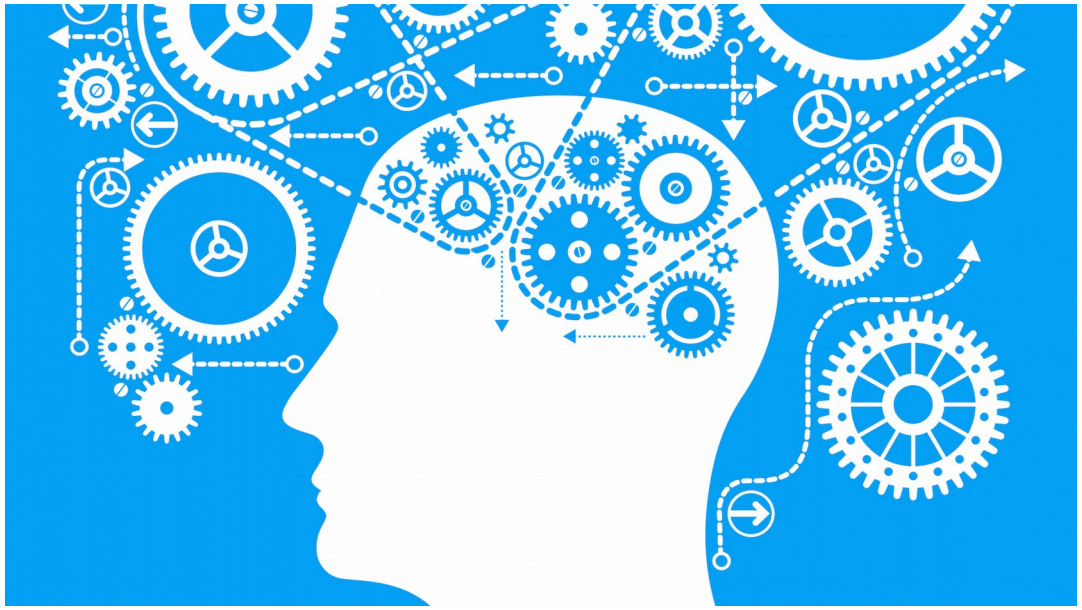
- Defined in a human context: able to
  - Reason in an abstract manner
  - Mental representation (purpose based?)
  - Problem solving
  - Learning
  - emotional knowledge
  - creativity
- *Most of all it is* ***adaptation to mee the demands of the environmen effectively***

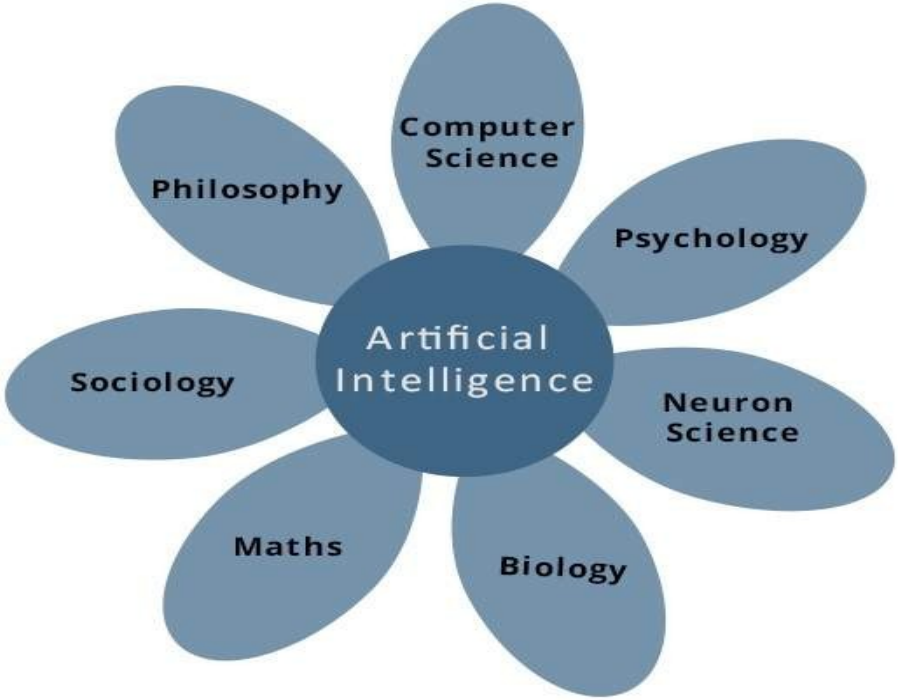




Artificial Intelligence

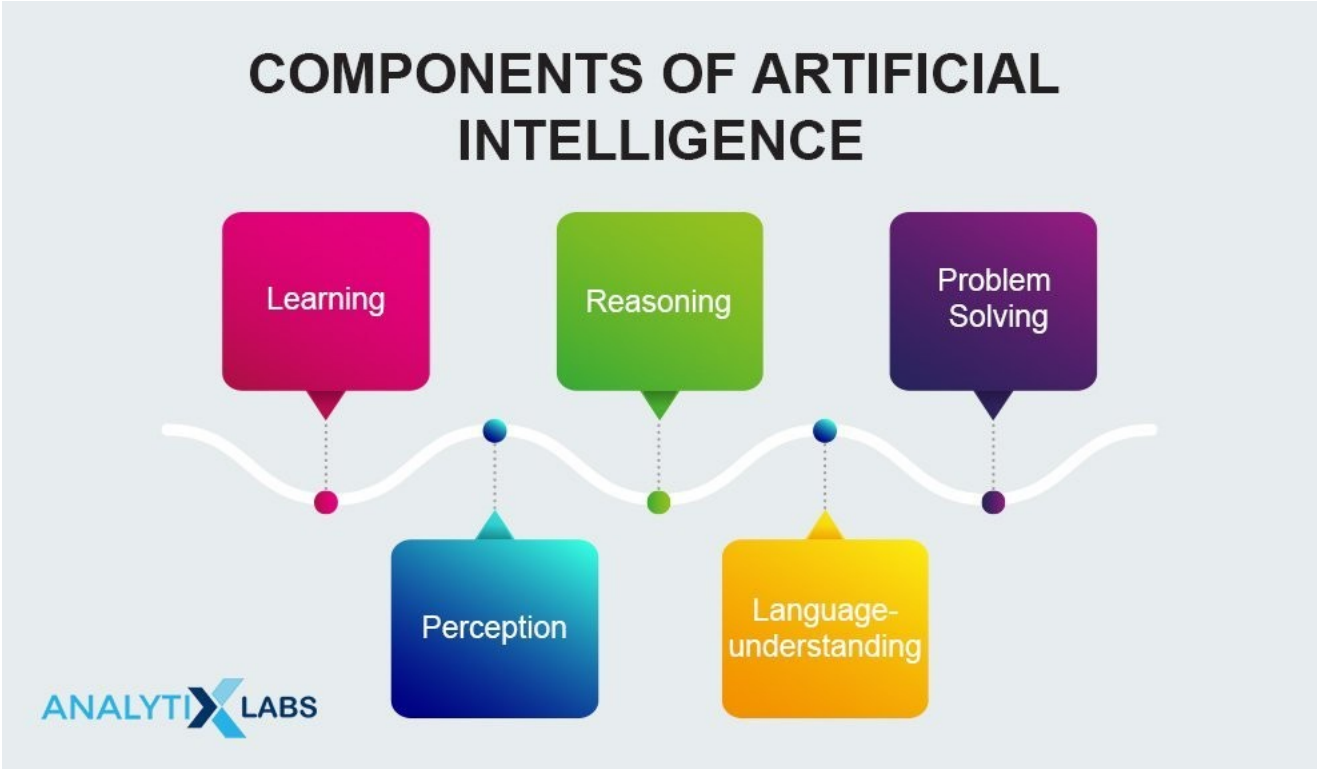
- As a discipline it attempts to understand and develop intelligent entities
- The fundamental assumptions made under AI are:
  - Workings of human mind can be explained in terms of computation
  - Computers can do the right thing given correct premises and reasoning rules
- It is an interdisciplinary field brought forth by the efforts of several disciplines

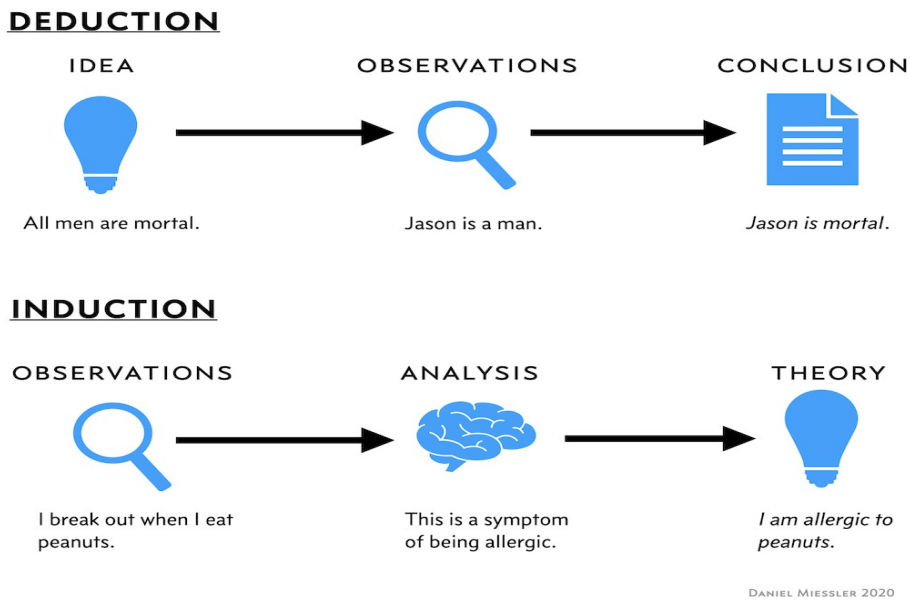




Components of an AI System

- **Perception**: scan the environment using different sensors – Camera, Ultrasonic sensors, LiDAR, etc.
  - Internally the scanned information is interpreted to understand the background
- **Language Understanding**: speech, written statement, signs and symbols
- **Learning**: trial and error based learning, memorizing solutions to different problems, vocabulary etc.
  - Use these later for different task
- **Reasoning**: make inferences given a set of premises/based on the current situation
  - Inductive vs Deductive inferences/reasoning
- **Problem Solving**: Given a data find a solution to a problem
  - Special purpose solution: customized solution to a specific problem
  - General purpose solution: solution to a

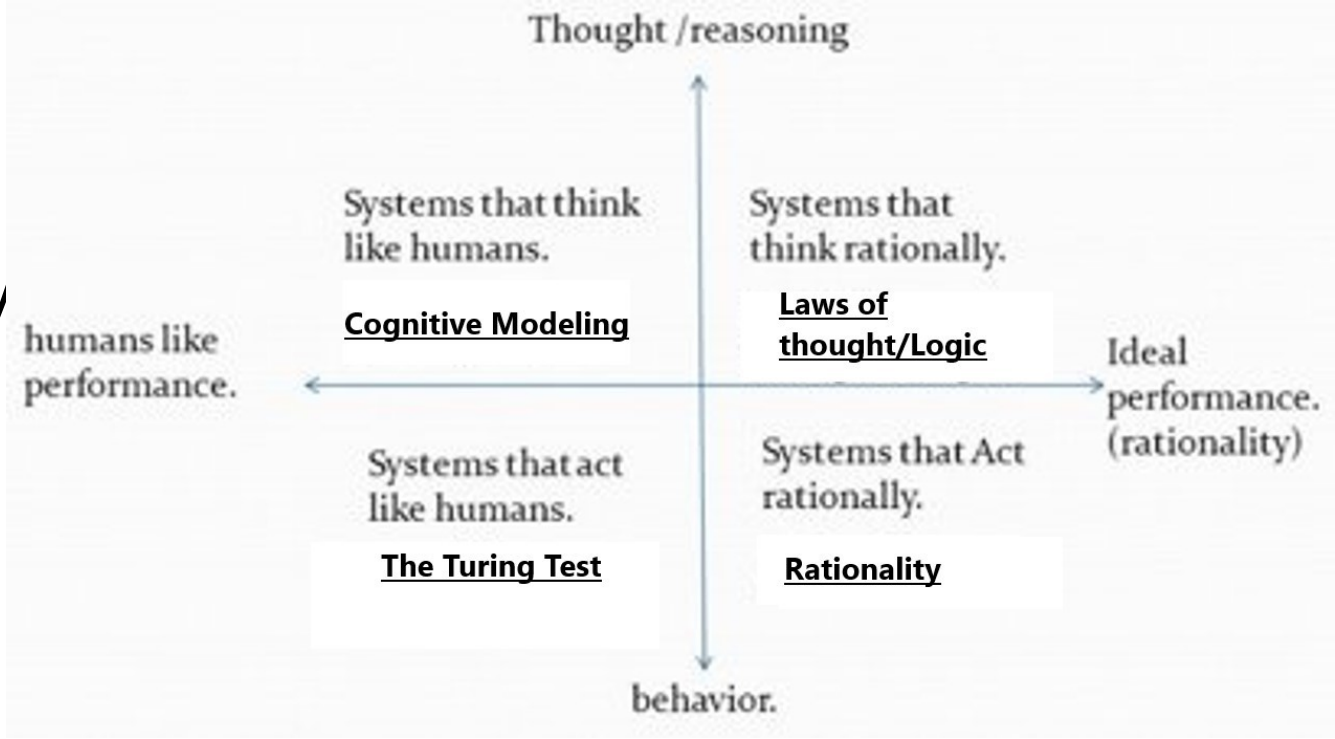




Approaches to AI

- Several approaches exist to study AI:
  - Human Centric approach: which derives much from humans
    - Human thinking – internal
    - Human Action – external

  - Dissociate from humans and focus on doing the 'right thing' – (perfection – because humans are limited)
    - Rational thinking - process
    - Rational action – behavior
    - *Important because there are numerous contexts where mimicking human thought/action is not important*



Human-centered Approach to AI

- Thinking like Humans:

  - Is a cognitive (?) modelling approach to AI
  - Requires understanding of the *internal* activities of the brain via
    - Introspection – observing ourselves
    - **Brain imaging** — observing the brain in action
    - Psychological experiments (E.g. Rorschach Test,)
  - Formulate a precise theory of the mind, it becomes possible to express the theory as a computer program.
  - Goal is to make program's input-output behavior match human behavior (After testing)

https://openpsychometrics.org/tests/HEMCR/



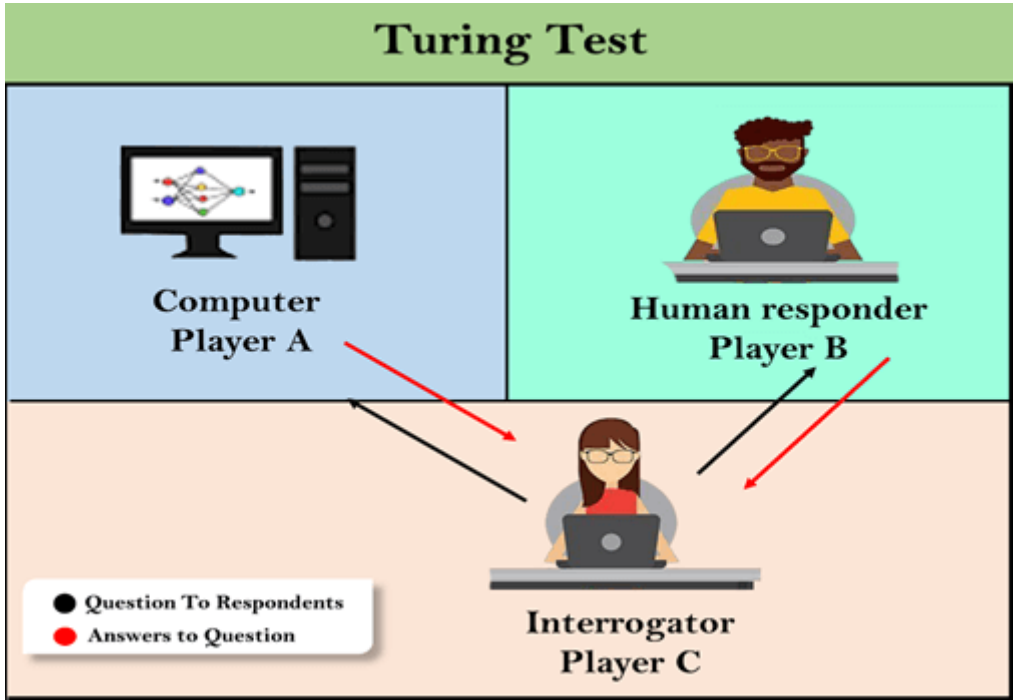


Human-centered Approach to AI

- Acting Like Humans (Blackbox):
  - A.K.A the Turing Test Approach
  - **Turing test**: Assume there's a person asking questions blindly (person/machine)
    - If the person cannot distinguish the machine's answer from the person's the machine has passed the test
    - Pragmatic/realistic/practical approach to answer the question of whether machines think or not
  - The machine should have the following:
    - Natural language processing
    - Knowledge representation
    - Automated reasoning
    - Machine learning
    - Computer Vision (Total Turing test)
    - Robotics (Total Turing Test)



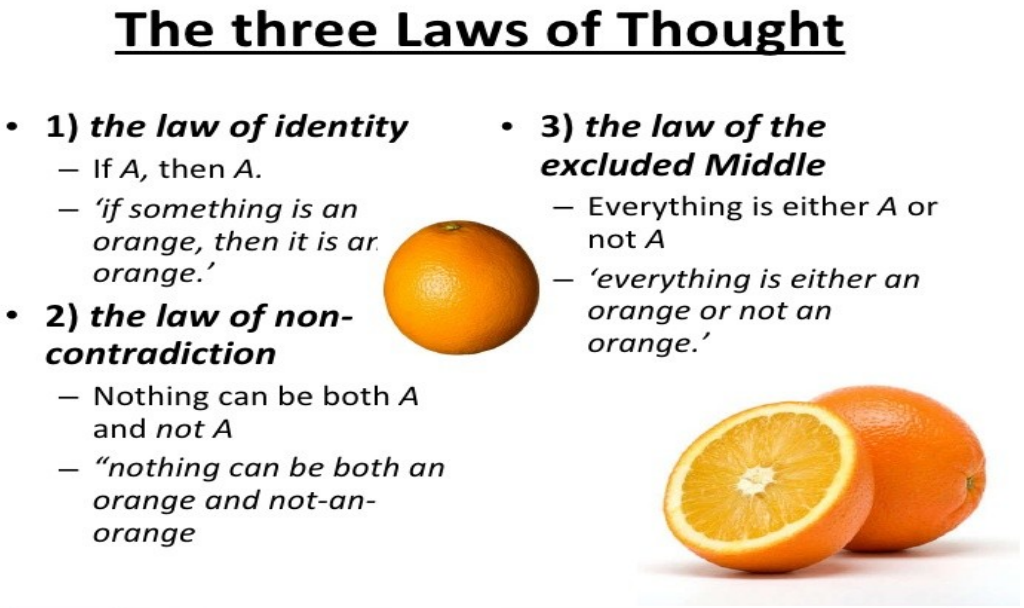
The Imitation Game

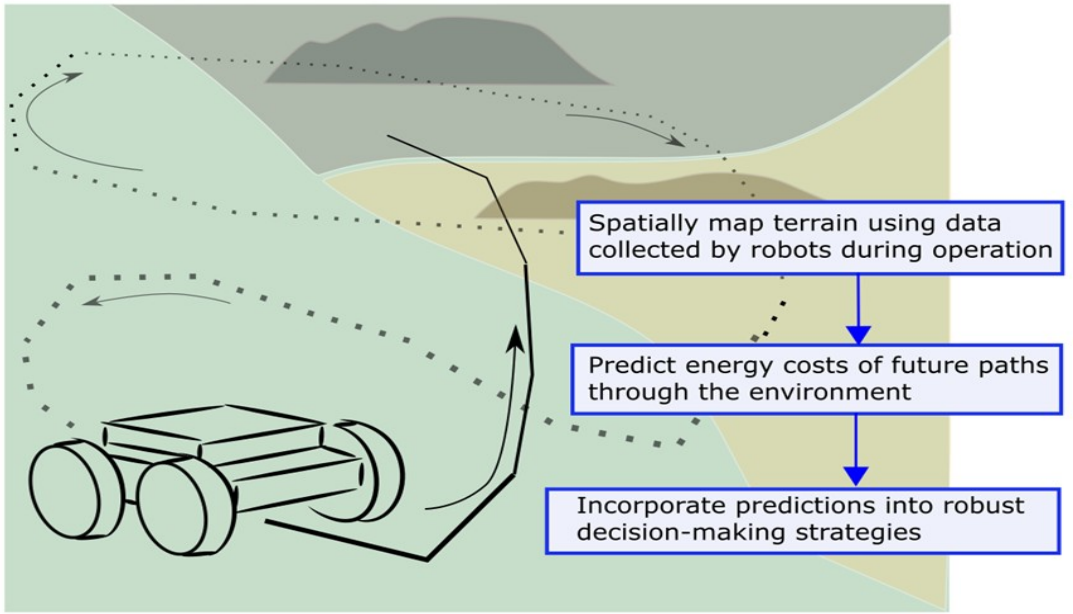


The **Rationalistic Approach** Approach to AI

Rationalistic Thinking:

- The “laws of thought” (Logic) approach
  - Aristotle attempted to codify an irrefutable reasoning processes.
  - His **syllogisms** provided patterns for argument structures that always yielded correct conclusions when given correct premises
- The Logic approach assumes a certain deterministic world – does not incorporate uncertainty (stochastic nature)
- The theory of **probability** fills this gap, allowing rigorous reasoning with uncertain information

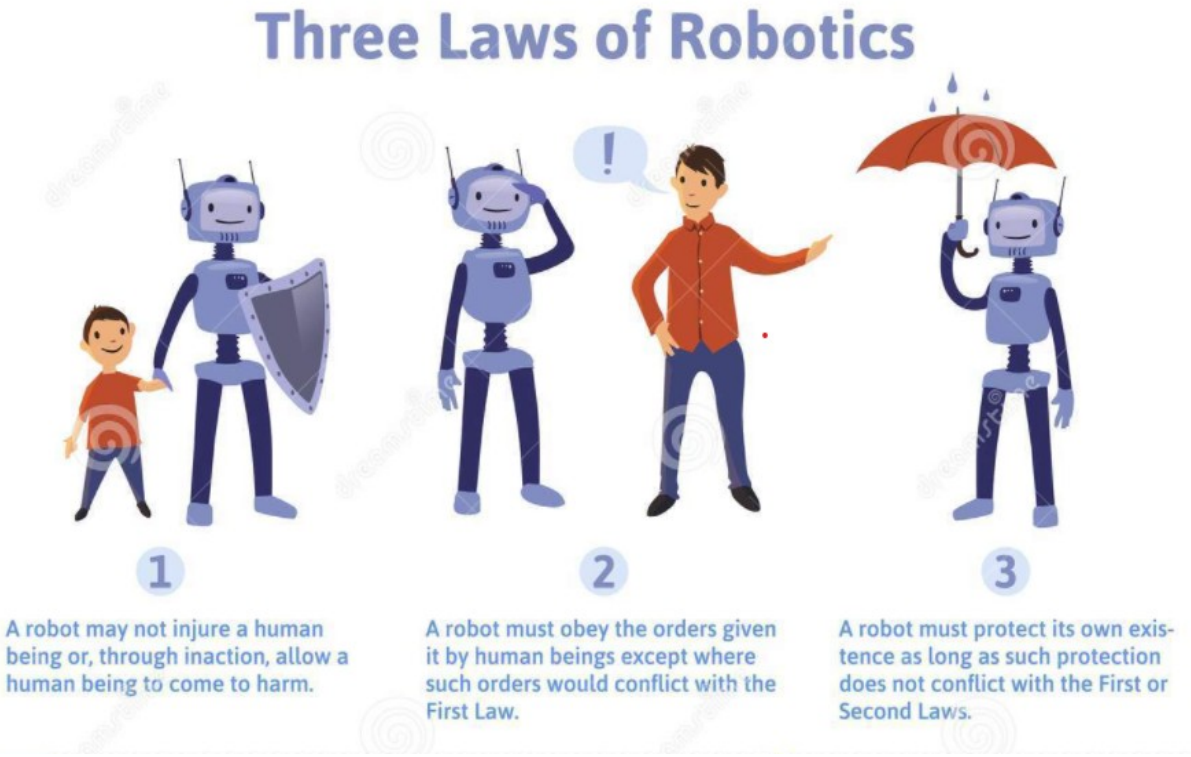




The **Rationalistic Approach** Approach to AI

Rationalistic Action (for systems that operate and act autonomously**):**

- A.K.A. the Rational Agent Approach
- An **agent** is just something that acts (*agent* comes from the Latin *agere*, to do)
- A **rational agent** is one that acts so as to achieve the best outcome or,
- Best outcome is defined by the objective given to the agent (Utility based)
- Rationality in some context's may not be limited to making inferences as in Logic
  - E.g. recoiling from a hot stove is a reflex action
- Another illustration is learning not only for erudition, but also because it improves our ability to generate effective behavior, especially in circumstances that are new (improvise)





Vs

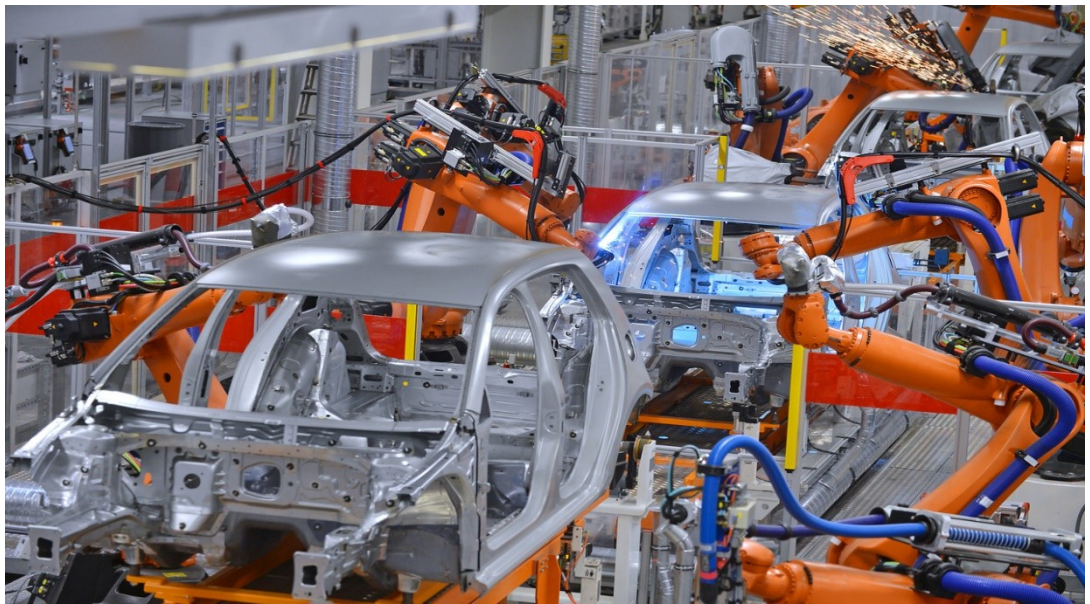


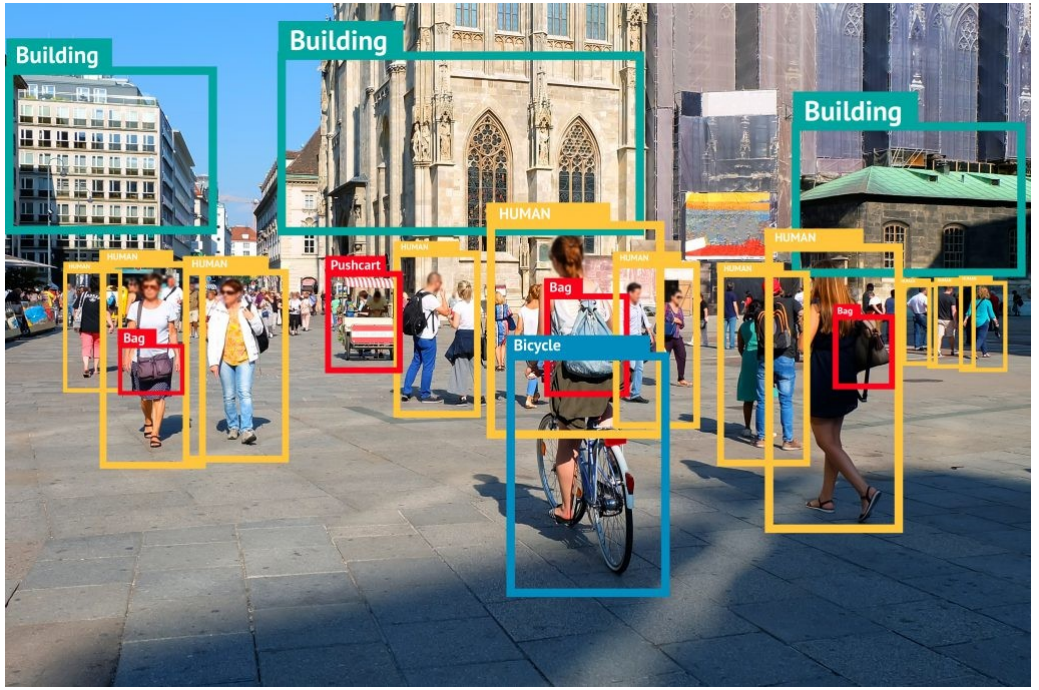
Application Areas of AI

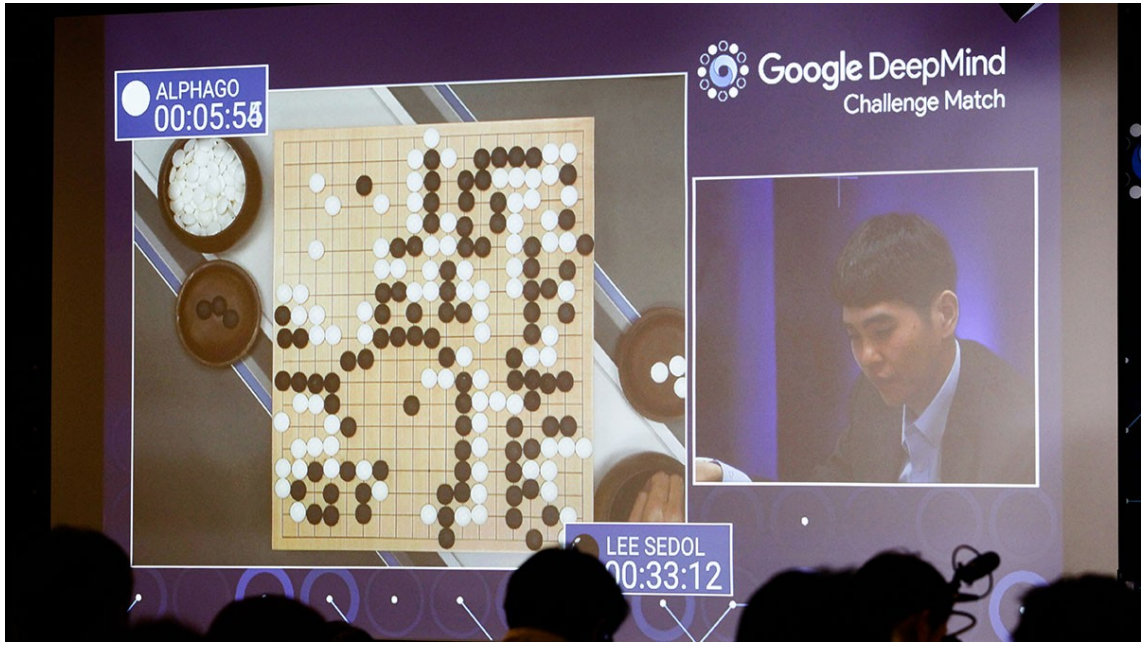
- Automating both the *mundane* and *expert* tasks

- Paradoxically, it turns out that it is the mundane tasks that are generally much the hardest to automate

  - Mundane Tasks (Image and Pattern Recognition)
    - Robots in Assembly plants etc.
  - Natural Language Processing (understanding, generation, translation)
  - Formal Tasks (Games, Puzzles, Mathematics)
  - Expert Tasks (Medical Diagnosis, Fault finding, Engineering)

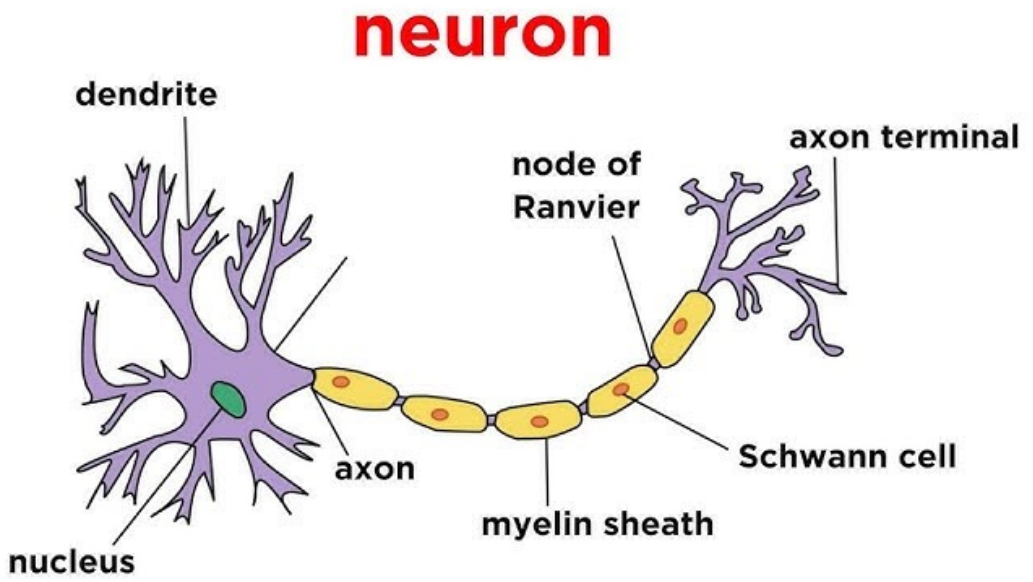




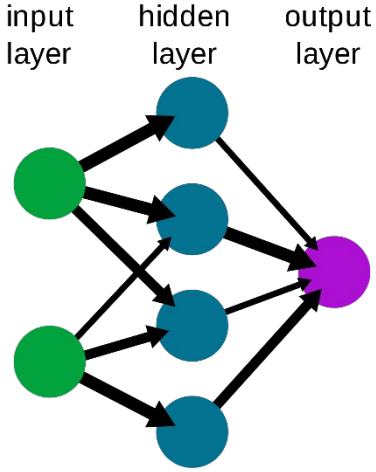


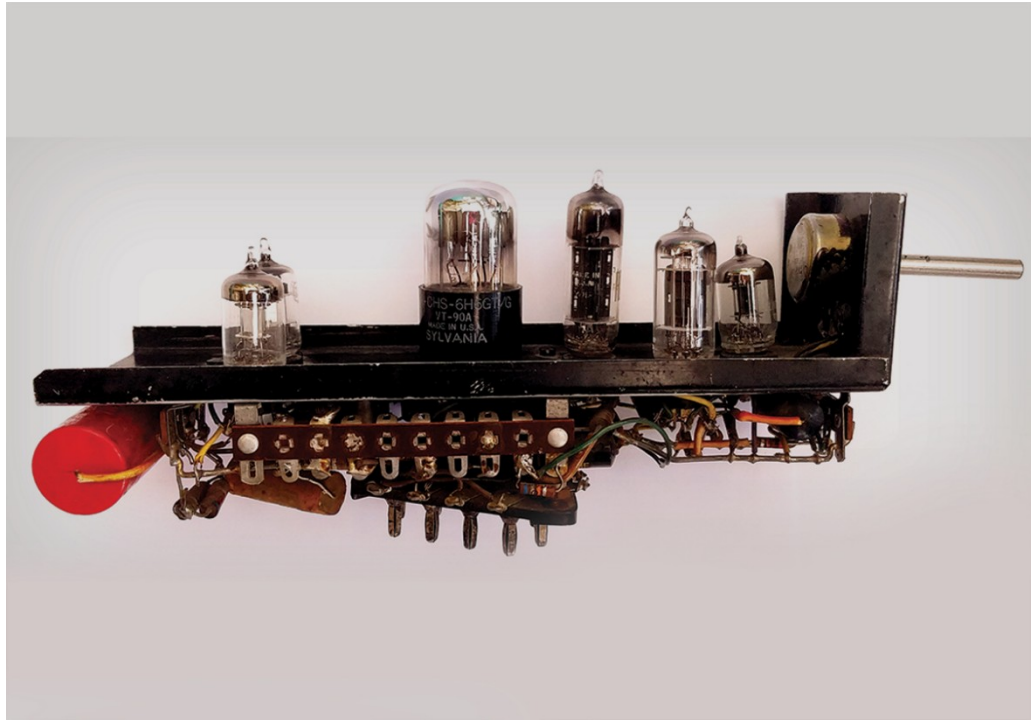
History of AI

- *The inception of artificial intelligence (1943–1956)*
  - First work by Warren McCulloch and Walter Pitts (1943).
  - They proposed a model of artificial neurons in which
    each neuron is characterized as being “on” or “off,”
  - Neuron’s switches state to “on” in response to stimulation by a sufficient number of neighboring neurons
  - Two undergraduate students at Harvard, Marvin Minsky and Dean Edmonds, built the first neural network computer in 1950



A simple neural network





History of AI - Continued

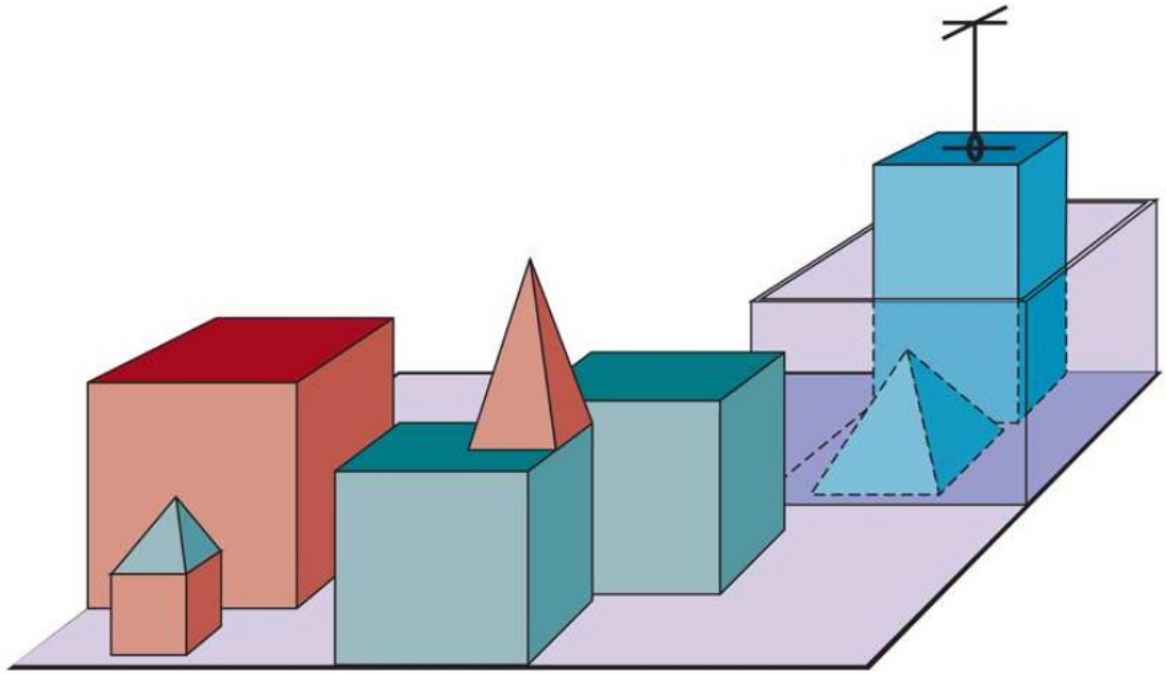
- *The inception of artificial intelligence (1943–1956)*
  - Alan Turing published the article “Computing Machinery and Intelligence” introducing *the Turing test, machine learning, genetic algorithms, and reinforcement learning*
- *Early enthusiasm, great expectations (1952–1969)*
  - Herbert Gelernter (1959) constructed the Geometry Theorem Prover, which was able to prove theorems that many students of mathematics would find quite tricky
  - Minsky supervised a series of students who chose limited problems/domains (microworlds)that appeared to require intelligence to solve.
  - Blocks world: a microworld where typical task in this world is to rearrange the blocks in a certain way, using a robot hand that can pick up one block at a time

I.—COMPUTING MACHINERY AND INTELLIGENCE

By A. M. Turing

1. *The Imitation Game.*

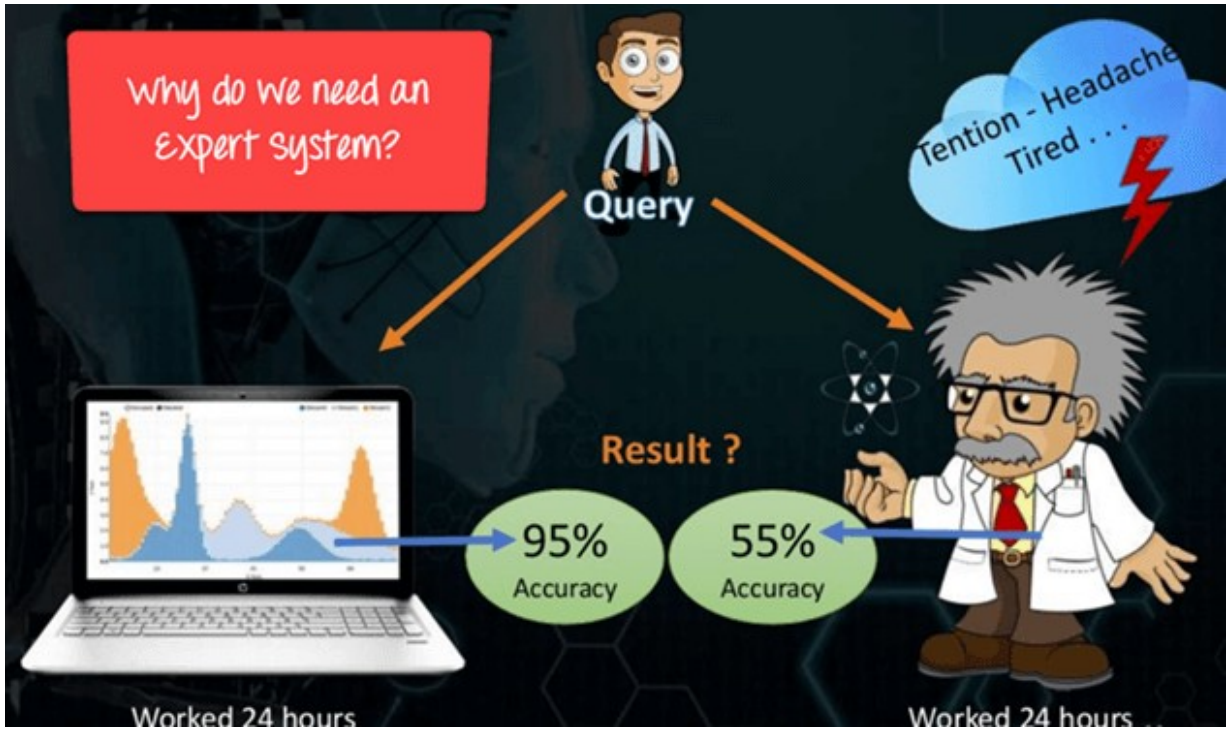
I propose to consider the question, ‘Can machines think?’ This should begin with definitions of the meaning of the terms ‘machine’ and ‘think’. The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words ‘machine’ and ‘think’ are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning and the answer to the question, ‘Can machines think?’ is to be sought in a statistical survey such as a Gallup poll. But this is absurd. Instead of attempting such a definition I shall replace the question by another, which is closely related to it and is expressed



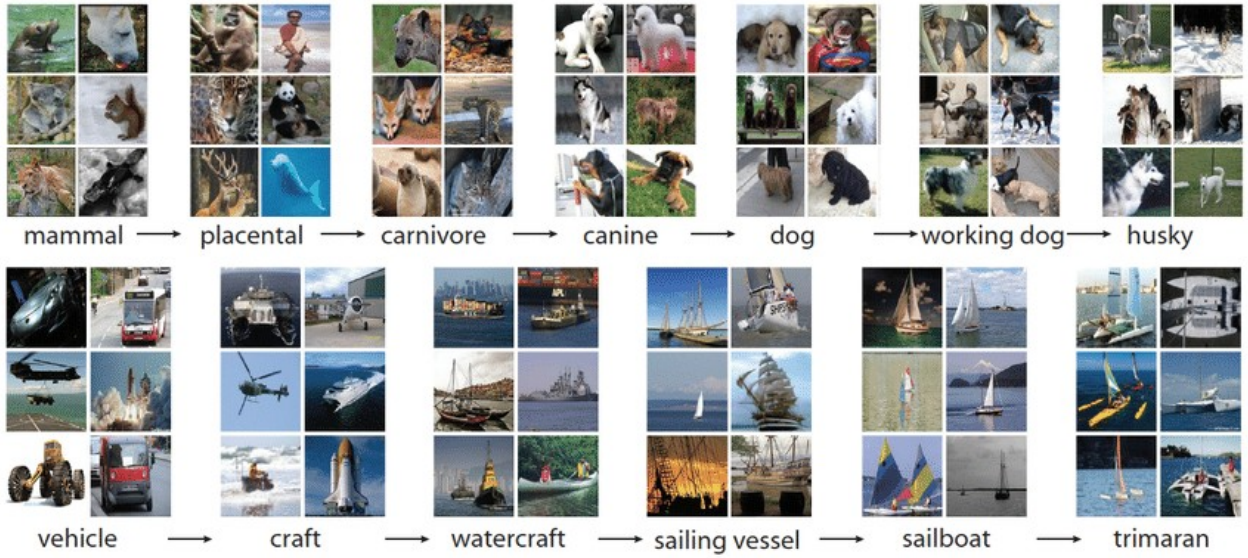
A scene from the blocks world. SHRDLU (Winograd, 1972) has just completed the command “Find a block which is taller than the one you are holding and put it in the box.”

History of AI - Continued

- *Expert systems (1969–1986)*
  - Expert system is a computer system emulating the decision-making ability of a human expert.
  - Expert systems are designed to solve complex problems by reasoning through bodies of knowledge, represented mainly as if–then rules
  - E.g. MYCIN system (1970s) used for diagnosing blood infections. With about 450 rules, MYCIN was able to perform as well as some experts

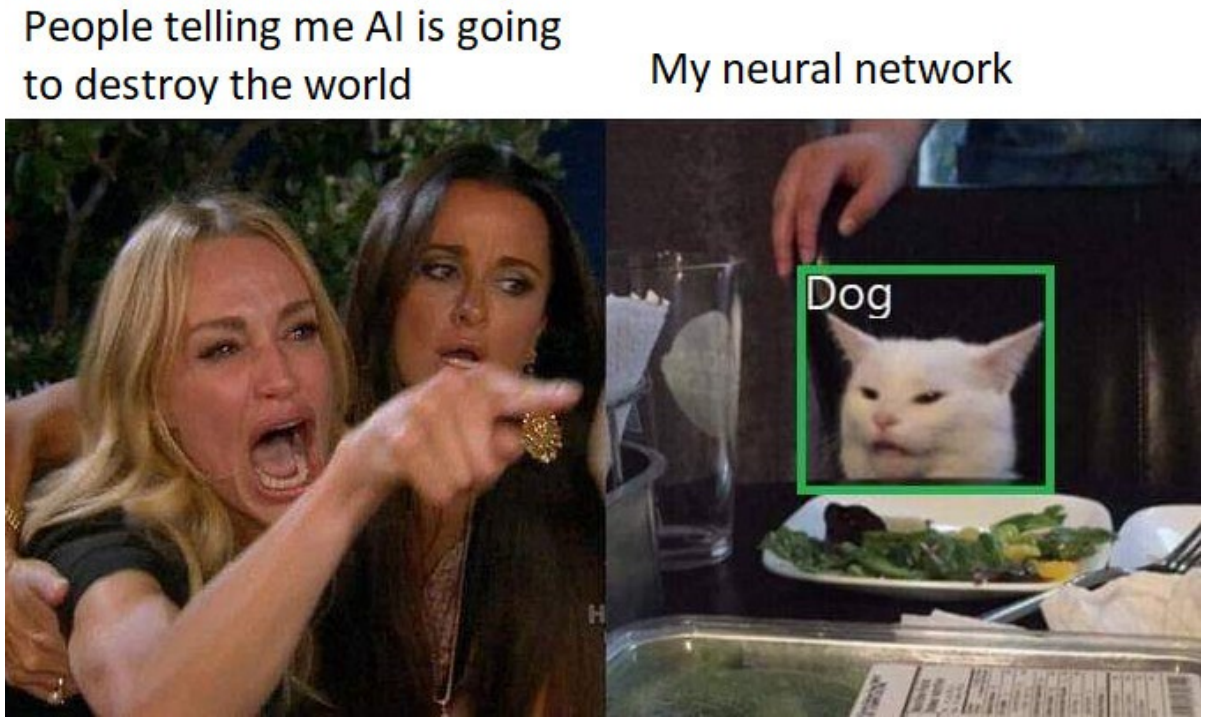


- Deep learning (2011–present):
  - Deep learning refers to machine learning using multiple layers of simple, adjustable computing elements (e.g. Neurons)
  - In the form of convolutional neural networks they found some success in handwritten digit recognition in the 1990s
  - ImageNet competition which used of 14 million images to perform a classification task in to 1000 categories – 63% accuracy achieved (2012)



What is the ultimate goal of AI?

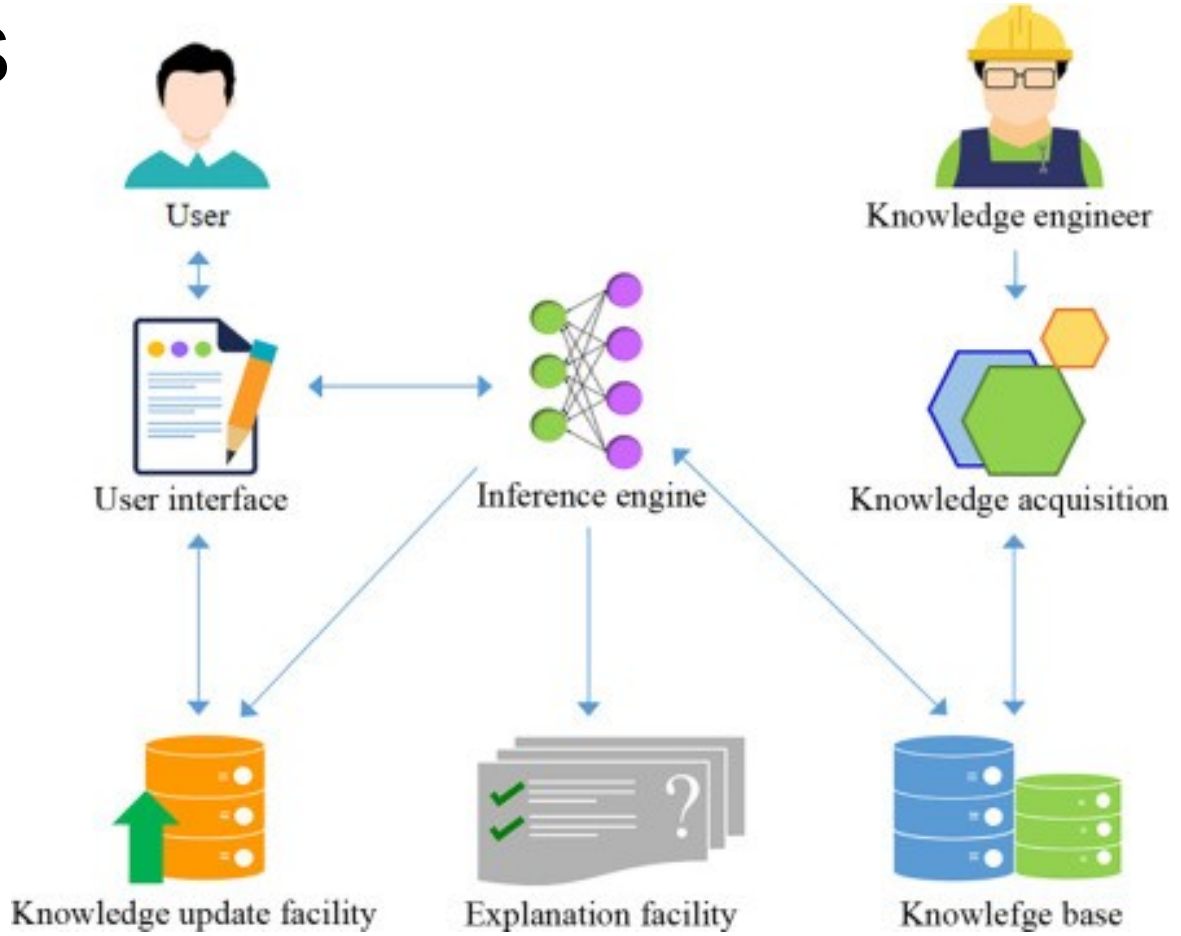
- Two major factions exist concerning the ultimate goal of AI
  - Weak AI Advocates:
    - If people need to make a computer that thinks, they should replace the circuits with proteins.
    - Generally, able to make robust computer systems that mostly provide decision support and decision making
    - Work on limited set of problems
  - Strong AI Advocates:
    - In the future technologists will one day be able to build a computer with all the cognitive, memory, and emotional capabilities of the human brain.
    - They believe that computers will someday be more intelligent than human beings
    - Work on diverse range of problems



Knowledge based-Systems

- Systems that use a knowledge base to make inferences for the purpose of problem solving
    - Knowledge base: storage mechanism for complex information
        - Consisted of facts about certain discipline or a topic at hand
    - Inference Engine/Reasoning System: inferring/deducing new knowledge from the KB using if-then rules

- Knowledge base is different from regular databases (relational) as it has:
    - Pointer linked data in several chains (both forward and backward)
    - This in turn provides something that looks like an object model (classes, subclasses, instances)



Reading Assignment

1.Compare and contrast the difference and similarities between human intelligence and artificial intelligence.
2.Define in your own words: (a) intelligence, (b) artificial intelligence, (c) agent, (d) rationality, (e) logical reasoning
3.Is AI a science, or is it engineering? Or neither or both? Explain.